



Water Potability Classification Using Machine Learning

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1. Problem Statement:

Access to safe and clean drinking water is a fundamental human right and a critical public health concern. In many regions, including Saudi Arabia, ensuring water potability is a key priority aligned with Vision 2030's sustainability and quality of life goals. Traditional water quality testing relies on laboratory analysis, which is time consuming, expensive, and not always accessible in remote areas. This project addresses the challenge of automating water potability classification by leveraging physicochemical properties of water such as pH, hardness, chloramines, and turbidity to predict whether a water sample is safe for human consumption. By applying machine learning techniques to measurable water attributes, this project aims to provide a fast, scalable, and cost-effective solution for water safety assessment.

2. Project Aim & Objectives:

2.1. Project Aim:

The aim of this project is to develop and evaluate machine learning classification models capable of predicting water potability based on physicochemical properties, supporting data driven decisions for safe drinking water assessment in alignment with Saudi Vision 2030 sustainability goals.

2.2. Project Objectives:

- a) To analyse and preprocess the Water Potability dataset by identifying and handling missing values, outliers, and class imbalance across physicochemical features.
- b) To perform Exploratory Data Analysis (EDA) to uncover patterns and relationships between water quality attributes and potability.
- c) To build four machine learning classification models (Decision Tree, Random Forest, KNN, SVM) to classify water samples as potable or non-potable.
- d) To evaluate and compare model performance using metrics including accuracy, precision, recall, and F1-score, with analysis of each model's confusion matrix.
- e) To improve model performance by applying data scaling techniques to the worst performing model and optimizing hyperparameters via GridSearchCV.
- f) To deploy the best-performing model as an interactive web application using Streamlit, enabling real-time water potability prediction.



3. Methodology:

This project follows the CRISP-DM (Cross Industry Standard Process for Data Mining) methodology^[6], which provides a structured framework for planning and executing data mining projects. The six phases are applied to this project as follows:

- a) **Phase 1 Business Understanding:** The goal is to automate water potability classification using physicochemical properties of water samples. Success is defined as building a reliable classification model that accurately differentiates potable water from non-potable water, reducing dependency on costly and time-consuming laboratory testing. This directly supports Saudi Vision 2030's commitment to environmental sustainability and public health^[5].
- b) **Phase 2 Data Understanding:** The Water Potability dataset is obtained from Kaggle, containing 3,276 water samples with 9 physicochemical features. Initial exploration is conducted to understand feature distributions, we detected missing values in pH, Sulfate, and Trihalomethanes columns, identify class imbalance in the target variable, and examine correlations between features.
- c) **Phase 3 Data Preparation:** Data preprocessing steps include handling missing values using mean imputation, detecting and treating outliers, and applying feature scaling to address the significantly different value ranges across features. AI ethics principles are incorporated to ensure fair and unbiased preprocessing throughout this phase.
- d) **Phase 4 Modeling:** Four classification models are built: Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Machine Vector (SVM). Each model is first trained using default parameters with an 80/20 train-test split, then optimized using GridSearchCV to identify the best hyperparameters for each model.
- e) **Phase 5 Evaluation:** Models are evaluated using accuracy, precision, recall, and F1-score. Confusion matrices are analyzed for each model, and performance comparisons are made before and after hyperparameter tuning. The worst-performing model is identified and scaling techniques are applied to assess improvement.
- f) **Phase 6 Deployment:** The best-performing model is deployed as an interactive web application using Streamlit, allowing users to input water quality measurements and instantly receive a potability prediction (potable or non-potable).



4. AI Ethics:

Developing a machine learning model for water safety assessment carries significant ethical responsibilities, as incorrect predictions can directly impact human health. In alignment with the Saudi Data and Artificial Intelligence Authority (SDAIA) AI Ethics Principles, the following ethical considerations are incorporated throughout this project:

- a) **Reliability and Safety:** Given that this model's predictions directly relate to human health and drinking water safety, reliability is of utmost importance. All three models are evaluated using multiple performance metrics and confusion matrices to ensure that the rate of false negatives (classifying unsafe water as potable) is minimized. A false negative in this context poses a direct public health risk and is therefore given special attention during evaluation.
- b) **Privacy and Data Protection:** The dataset used in this project contains no personally identifiable information as it consists solely of physicochemical measurements of water samples collected from water bodies, ensuring full compliance with data privacy principles and eliminating any risk of personal data exposure.
- c) **Transparency:** The Decision Tree model provides a fully interpretable visual structure, allowing stakeholders such as water treatment authorities to understand how classification decisions are made. The feature importance output of the Random Forest model further enhances transparency by identifying which water quality attributes most influence potability predictions.
- d) **Human Oversight:** The deployed Streamlit application is designed as a decision support tool, not a replacement for certified water quality testing. Final potability assessments must remain under the authority of qualified environmental and public health professionals, with the model serving as a rapid preliminary screening tool to guide further testing.



5. Data Description

The dataset used in this project is the Water Potability Dataset we sourced from Kaggle. The dataset contains physicochemical measurements of water samples collected from multiple water bodies and is used to determine whether a water sample is safe for human consumption. The dataset consists of 3,276 rows and 10 columns. Missing values are present in three features: pH (491 missing), Sulfate (781 missing), and Trihalomethanes (162 missing). The dataset contains all features in **Table 5.1**:

	Feature	Description	Unit
1	pH	Measures acidity or alkalinity of water	0–14 scale
2	Hardness	Measure of mineral content caused by calcium and magnesium salts	mg/L
3	Solids	Total dissolved solids in water	Ppm
4	Chloramines	Amount of chloramines used as disinfectants in water	Ppm
5	Sulfate	Amount of sulfate dissolved in water	mg/L
6	Conductivity	Electrical conductivity of water	μS/cm
7	Organic_carbon	Amount of organic carbon from natural and synthetic sources	ppm
8	Trihalomethanes	Chemical compounds found in chlorinated water	μg/L
9	Turbidity	Measure of light emitting properties of water	NTU
10	Potability	Whether water is safe for human consumption 1 (Potable) and 0 (Not Potable)	Target

Table 5.1. all features in the dataset.

The target variable is Potability, a binary label indicating whether a water sample is safe for drinking. The dataset contains a class imbalance with approximately 2000 non-potable samples and 1,250 potable samples, which is addressed during the preprocessing phase.

6.Exploratory Data Analysis:

Before building machine learning models, a thorough exploratory data analysis was conducted to understand the structure and quality of the Water Potability dataset.

- Initial Inspection:** The dataset contains 3,276 rows and 10 columns. All features are continuous numerical variables except for the binary target variable Potability. An initial inspection confirmed that no duplicate rows exist in the dataset.
- Missing Values:** Three features contain missing values: pH (491 missing), Sulfate (781 missing), and Trihalomethanes (162 missing). Missing values were handled by imputing the mean value of each respective feature, preserving the overall data distribution without introducing bias.
- Outliers:** Box plots were used to detect outliers across all features. Several features including Solids, Conductivity, and Trihalomethanes showed the presence of outliers as shown in **Figure 6.1**. These were treated using the Interquartile Range (IQR) method, where values falling below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ were capped to the respective boundary values as shown in **Figure 6.2**.

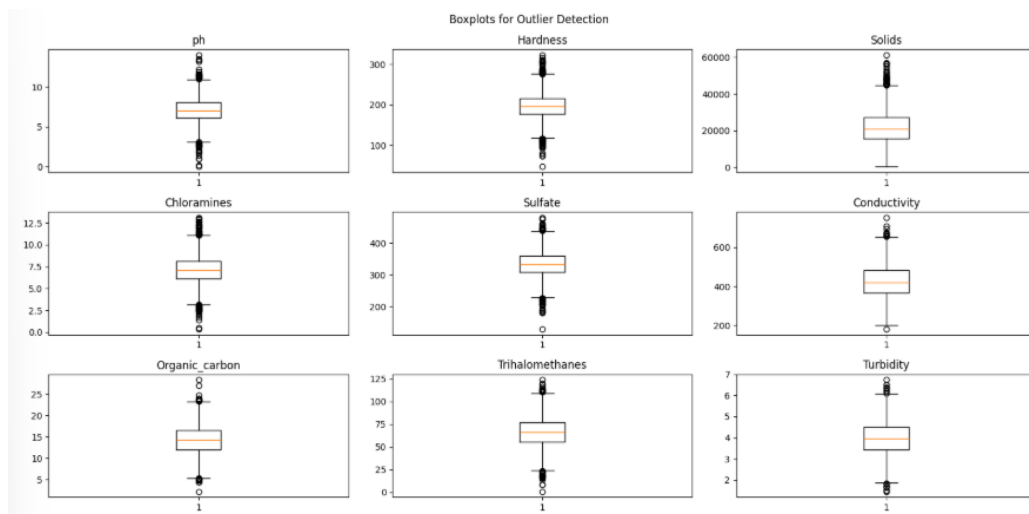


Figure 6.1. Boxplots Before Handling Outliers.

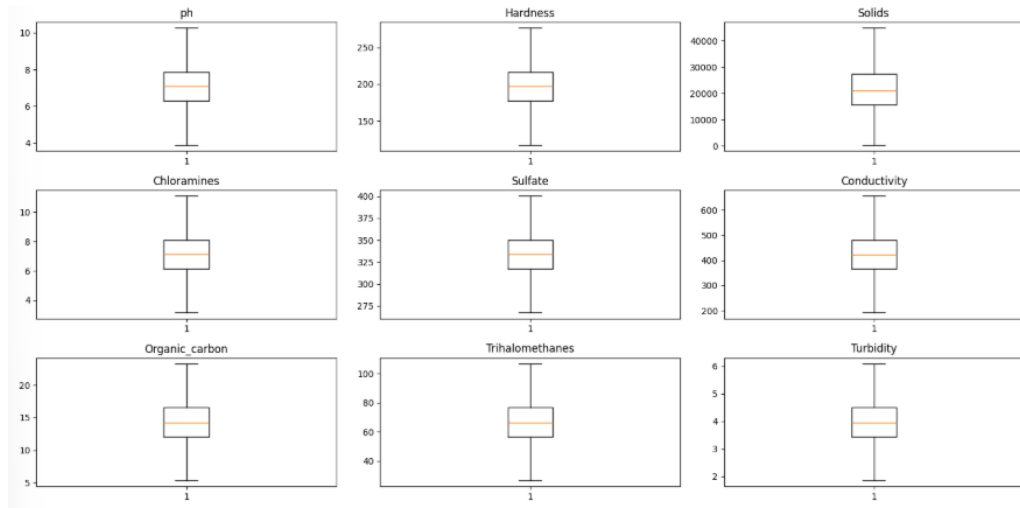


Figure 6.2. Boxplots After Handling Outliers.

- d. Class Distribution: The target variable Potability is imbalanced, with almost 2000 non-potable samples and 1,250 potable samples as you can see in **Figure 6.3**. This imbalance is solved using SMOTE as shown in in **Figure 6.4**.

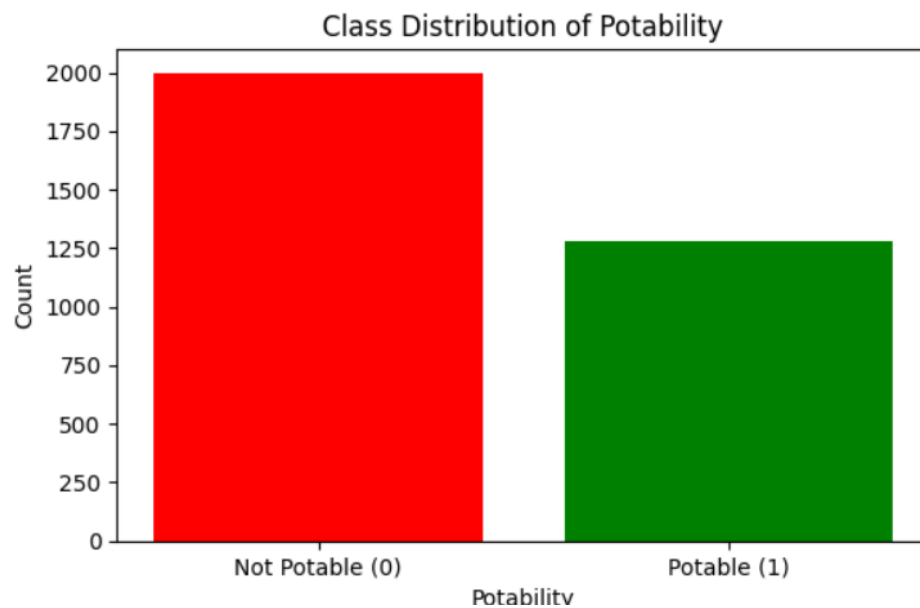


Figure 6.3. Class Distribution of Potability.

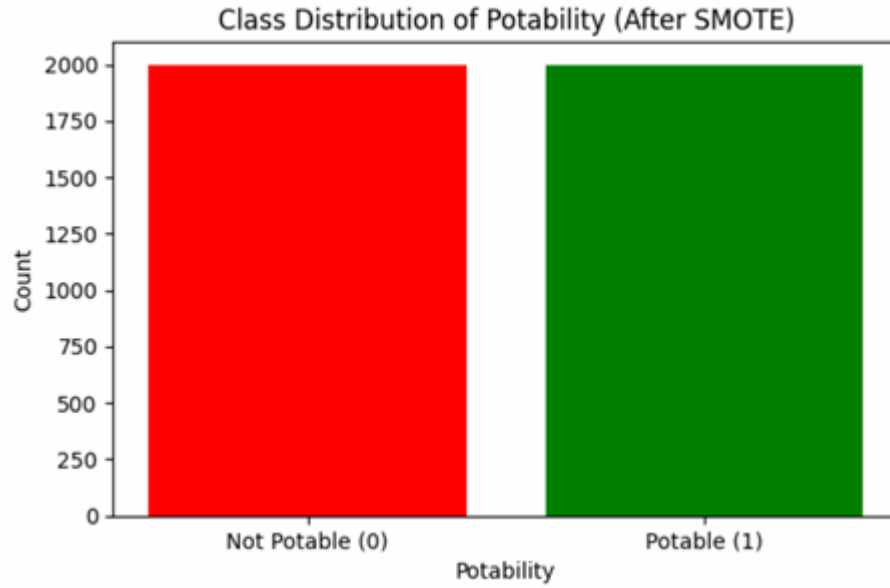


Figure 6.4. class Distribution after using SMOTE.

- e. Feature Distribution: Histograms were plotted for all features to examine their distributions as you can see in **Figure 6.5**. Most features follow an approximately normal distribution, while Solids shows a right skewed distribution, indicating the presence of high dissolved solid concentrations in some samples.

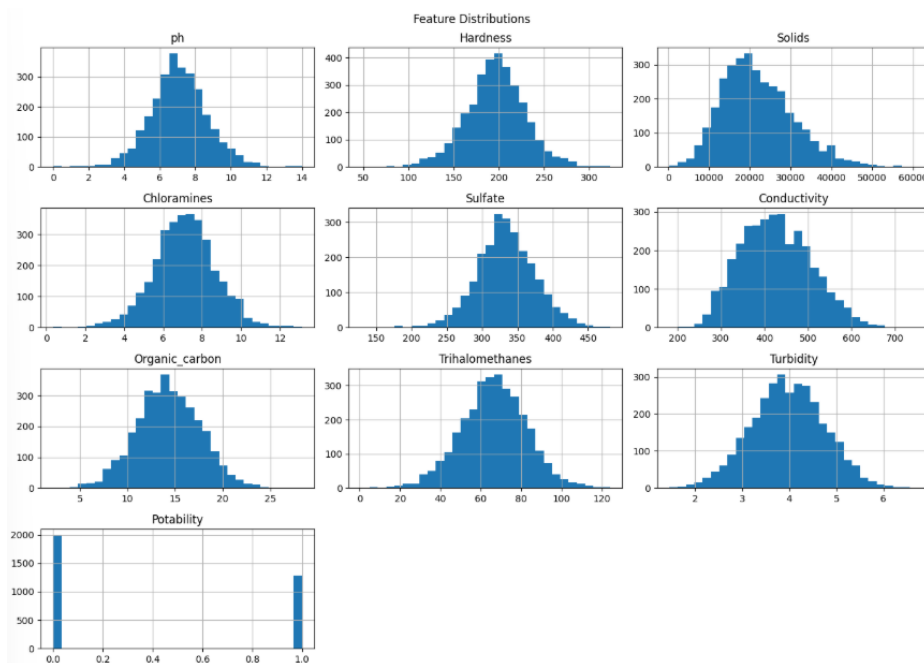


Figure 6.5. Histograms of all the features.



- f. Correlation Analysis: The Correlation Matrix revealed no strong linear correlations between features and the target variable as you can see in **Figure 6.6**, confirming that non-linear models such as Decision Tree and Random Forest are more appropriate for this dataset than linear models.

	ph	Hardness	Solids	Chloramines	Sulfate	\
ph	1.000000	0.082096	-0.089288	-0.034350	0.018203	
Hardness	0.082096	1.000000	-0.046899	-0.030054	-0.106923	
Solids	-0.089288	-0.046899	1.000000	-0.070148	-0.171804	
Chloramines	-0.034350	-0.030054	-0.070148	1.000000	0.027244	
Sulfate	0.018203	-0.106923	-0.171804	0.027244	1.000000	
Conductivity	0.018614	-0.023915	0.013831	-0.020486	-0.016121	
Organic_carbon	0.043503	0.003610	0.010242	-0.012653	0.030831	
Trihalomethanes	0.003354	-0.013013	0.009143	0.017084	-0.030274	
Turbidity	-0.039057	-0.014449	0.019546	0.002363	-0.011187	
Potability	-0.003556	-0.013837	0.033743	0.023779	-0.023577	

	Conductivity	Organic_carbon	Trihalomethanes	Turbidity	\
ph	0.018614	0.043503	0.003354	-0.039057	
Hardness	-0.023915	0.003610	-0.013013	-0.014449	
Solids	0.013831	0.010242	-0.009143	0.019546	
Chloramines	-0.020486	-0.012653	0.017084	0.002363	
Sulfate	-0.016121	0.030831	-0.030274	-0.011187	
Conductivity	1.000000	0.020966	0.001285	0.005798	
Organic_carbon	0.020966	1.000000	-0.013274	-0.027308	
Trihalomethanes	0.001285	-0.013274	1.000000	-0.022145	
Turbidity	0.005798	-0.027308	-0.022145	1.000000	
Potability	-0.008128	-0.030001	0.007130	0.001581	

	Potability
ph	-0.003556
Hardness	-0.013837
Solids	0.033743
Chloramines	0.023779
Sulfate	-0.023577
Conductivity	-0.008128
Organic_carbon	-0.030001
Trihalomethanes	0.007130
Turbidity	0.001581
Potability	1.000000

Figure 6.6. The Correlation Matrix.

After completing all preprocessing steps, **Figure 6.7** shows the first 10 rows of our cleaned dataset.

First 10 rows of preprocessed data:

	ph	Hardness	Solids	Chloramines	Sulfate	Conductivity	Organic_carbon	Trihalomethanes	Turbidity
0	7.080795	204.890455	20791.318981	7.300212	368.516441	564.308654	10.379783	86.990970	2.963135
1	3.889107	129.422921	18630.057858	6.635246	333.775777	592.885359	15.180013	56.329076	4.500656
2	8.099124	224.236259	19909.541732	9.275884	333.775777	418.606213	16.868637	66.420093	3.055934
3	8.316766	214.373394	22018.417441	8.059332	356.886136	363.266516	18.436524	100.341674	4.628771
4	9.092223	181.101509	17978.986339	6.546600	310.135738	398.410813	11.558279	31.997993	4.075075
5	5.584087	188.313324	28748.687739	7.544869	326.678363	280.467916	8.399735	54.917862	2.559708
6	10.223862	248.071735	28749.716544	7.513408	393.663396	283.651634	13.789695	84.603556	2.672989
7	8.635849	203.361523	13672.091764	4.563009	303.309771	474.607645	12.363817	62.798309	4.401425
8	7.080795	118.988579	14285.583854	7.804174	268.646941	389.375566	12.706049	53.928846	3.595017
9	10.258615	227.231469	25484.508491	9.077200	400.322434	563.885481	17.927806	71.976601	4.370562

Figure 6.7. Sample of the data after preprocessing.



7. Model Selection:

Based on the exploratory data analysis (EDA) conducted in Section 2, four machine learning models were selected: Decision Tree, Random Forest, KNN, SVM. The choice of these models is justified by the nature of the dataset and the patterns revealed during EDA. The correlation heatmap indicated no strong linear relationships between the physicochemical features and the target variable (Potability), suggesting that nonlinear models would be more effective for classification. Decision Tree and Random Forest are well-suited for such data because they can capture complex, nonlinear interactions between features without requiring prior assumptions about data distribution.

The Decision Tree model was chosen for its interpretability and transparency, allowing stakeholders to visualize how each feature contributes to the classification decision^[1]. The Random Forest model, an ensemble of multiple decision trees, was selected for its robustness and ability to reduce overfitting while improving generalization performance^[2]. The KNN model was included as a distance-based classifier that can provide complementary insights by classifying samples based on similarity to neighboring data points^[3].

Previous research supports the effectiveness of these models for water potability prediction. Kadiwal (2021) demonstrated that tree-based classifiers such as Decision Tree and Random Forest achieve reliable accuracy^[4]. Similarly, Sharma et al. (2022) and Patel et al. (2023) found that Random Forest and KNN models perform well in predicting drinking water safety based on physicochemical attributes^{[5][6]}. Therefore, these models were selected to ensure a balance between interpretability, accuracy, and computational efficiency in assessing water safety.



8. Model Estimation:

The experiment was done on the water potability dataset, we built four models -KNN, SVM, decision tree, random forest- to predict if a new data point is safe for humans to drink or not. We cleaned and transformed the data beforehand to make it more reliable and representative of the real-world water samples; we filled missing values of columns with their means, and handled outliers using the IQR method, then we balanced the two classes using SMOTE. We split the training data in a 80:20 ratio, and we trained the four models on the training set. And it was evaluated using accuracy, confusion matrix and classification reports.

9. Model evaluation:

we evaluated the performance of our four classification models (Decision Tree, Random Forest, KNN, and SVM). We evaluated the models first using their default parameter values, and then optimized them using GridSearchCV (with 5-fold cross-validation).

a & b. Default Models and Hyperparameter Tuning: To improve the performance of our default models, we systematically tested different combinations of parameters to find the optimal values for each model:

- **Random Forest:** The best parameters found were max_depth: 20 and n_estimators: 100.
- **SVM:** We applied data scaling using MinMaxScaler prior to tuning to improve performance. The best parameters found were C: 1 and gamma: 1.
- **KNN:** The optimal number of neighbors found was n_neighbors: 3.
- **Decision Tree:** The best parameters were max_depth: 15 and min_samples_split: 2.

c. Performance Comparison: The results of the experiments are presented in Table 9.1 below, comparing the models using multiple metrics (Accuracy, Precision, Recall, and F1-score based on the weighted average) before and after hyperparameter tuning.

ML Models	Accuracy	Precision	Recall	F1-Score
Random Forest (Default)	69.25%	70%	70%	70%
Random Forest (Tuned)	71.13%	7%	71%	71%
SVM (Default)	51.25%	53%	51%	46%
SVM (Tuned and scaled)	68.14%	69%	68 %	63%
KNN (Default)	60.75%	61%	61%	61%
KNN(Tuned)	62.13%	62%	62 %	62%
Decision Tree (Default)	60%	60%	59%	59%
Decision Tree (Tuned)	59.38%	59%	59%	5%



Table 9.1 Comparison of Models' Performance (Default vs Tuned)

d. Underfitting and Overfitting Analysis:

- **Overfitting:** Both the Decision Tree and Random Forest models showed clear signs of overfitting in their default states, achieving a perfect training accuracy of 1.0 (100%) while their testing accuracies were significantly lower (60.00% and 69.25% respectively).
- **Underfitting:** The default SVM model suffered from underfitting, with a training accuracy of 51.88% and testing accuracy of 51.25%. This was successfully resolved by applying scaling and tuning, which brought the test accuracy up to 68.14%

10. Deployment:

The main python code for running the streamlit is called app.py, and water_potability_model (3).pkl is the random forest model file that was generated during joblib. As shown in **Figure 10.1** and **Figure 10.2** is the Streamlit UI we created to act as a bridge, as it takes raw user data via widgets, and it uses the .pkl file to generate an immediate classification without retraining the model, you will find the Water Potability Prediction app in this link : [Streamlit](#)

The figure displays two side-by-side screenshots of a Streamlit web application titled "Water Potability Prediction". Both screens have a dark theme and a header that says "Enter the water characteristics below to check if it's safe to drink or not". Each screen contains ten input fields with sliders and a "Predict Potability" button. The inputs are: pH Level, Hardness, Solids, Chloramines, Sulfate, Conductivity, Organic Carbon, Trihalomethanes, and Turbidity. In the left screenshot (Figure 10.1), the predicted result is "The water is not Potable (Unsafe to drink)" shown in a red box. In the right screenshot (Figure 10.2), the predicted result is "The water is Potable (Safe to drink)" shown in a green box.

Parameter	Figure 10.1 Value	Figure 10.2 Value
pH Level	14.00	7.00
Hardness	300.00	150.00
Solids	1200.00	600.00
Chloramines	3.00	3.00
Sulfate	300.00	300.00
Conductivity	400.00	400.00
Organic Carbon	4.00	4.00
Trihalomethanes	0.06	0.06
Turbidity	4.00	4.00

Figure 10.1. Negative prediction using StreamLit

Figure 10.2. Positive prediction using StreamLit

The widgets are input fields meant to take continuous numbers as input and with the “predict probability” button being an action button that starts the classification process. After the classification is done a message is shown under the button to showcase the result of the classification, green and red is used for the potable and not potable classifications respectively for an easier understanding of the interface. We achieved this result with the code shown in **Figure 10.3**.

```
1 import streamlit as st
2 import pandas as pd
3 import joblib
4
5 model = joblib.load('water_potability_model (3).pkl')
6 st.title(" Water Potability Prediction")
7 st.write("Enter the water characteristics below to check if it's safe to drink or not")
8
9 ph = st.number_input("pH Level", min_value=0.0, max_value=14.0, value=7.0)
10 hardness = st.number_input("Hardness", min_value=0.0, value=150.0)
11 solids = st.number_input("Solids", min_value=0.0, value=600.0)
12 chloramines = st.number_input("Chloramines", min_value=0.0, value=3.0)
13 sulfate = st.number_input("Sulfate", min_value=0.0, value=300.0)
14 conductivity = st.number_input("Conductivity", min_value=0.0, value=400.0)
15 organic_carbon = st.number_input("Organic Carbon", min_value=0.0, value=4.0)
16 trihalomethanes = st.number_input("Trihalomethanes", min_value=0.0, value=0.06)
17 turbidity = st.number_input("Turbidity", min_value=0.0, value=4.0)
18
19 if st.button("Predict Potability"):
20     input_data = pd.DataFrame([[ph, hardness, solids, chloramines, sulfate,
21                               conductivity, organic_carbon, trihalomethanes, turbidity]],
22                               columns=['ph', 'Hardness', 'Solids', 'Chloramines', 'Sulfate',
23                                       'Conductivity', 'Organic_carbon', 'Trihalomethanes', 'Turbidity'])
24
25     proba = model.predict_proba(input_data)[0][1]
26     prediction = 1 if proba >= 0.49 else 0
27
28     st.markdown("---")
29     if prediction == 1:
30         st.success(" The water is Potable (Safe to drink)")
31     else:
32         st.error(" The water is not Potable (Unsafe to drink)")
```

Figure 10.3. StreamLit code

Everything you need for the streamlit is in this link: a71607809-web/WaterPotabilityApp

11.Conclusion:

In this project, four machine learning models were used to classify water samples as potable or non-potable: Decision Tree, Random Forest, KNN, and SVM. Different preprocessing techniques were applied, including handling missing values, outlier handling, feature scaling, and SMOTE for balancing. The results showed that Random Forest achieved the best overall performance with the base model having an accuracy of 72%, while the tuned SVM model also achieved strong performance with an accuracy of 68% after scaling and hyperparameter tuning. The findings highlighted the importance of preprocessing techniques, especially scaling and balancing the dataset using SMOTE.



Overall, the project demonstrated how machine learning can support faster and more efficient water potability prediction. This can help authorities and environmental organizations make quicker decisions related to public health and safe drinking water and help in various future projects that might need assessments of water safety. For future work, the project can be improved by using larger and geographically varied datasets, testing more advanced machine learning models, and integrating real-time water sensor data to make predictions more accurate and practical.

12. References:

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